

MSDM5004 Spring 2026

Homework 1 Part II

Due Mar. 1

3. Find the least squares polynomial of degree 1 for the data in the table, and compute the least squares error ($E = \|Ax - b\|$).

x_i	3.4212	3.9279	5.1779	6.8251	11.8603
y_i	13.5339	15.1001	18.9670	23.6083	36.9070

4. Write a code using MATLAB and the built-in function `svd( )` to determine the least squares polynomial of degree 2 of the following data

x_i	1.0700	1.2270	1.5410	1.8550	2.4830	2.7970
y_i	1.8547	1.9347	2.1013	2.2613	2.5880	2.7480